

Bakare Oluwatobiloba Olukayode

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Profile

Mechatronics Engineering graduate with hands-on experience in robotics, embedded systems, and data automation. Skilled in ROS2, Python, and C++ for robotic simulation, control, and industrial problem solving. Proven ability to design systems, automate workflows, and translate engineering concepts into functional prototypes.

Education

BELLS UNIVERSITY OF TECHNOLOGY, OGUN STATE | SEPTEMBER 2020 - JULY 2025

B.Eng. Mechatronics Engineering | CGPA: 4.42/5.00

Technical Highlights

- Simulated and controlled a 7-DOF robotic arm using ROS2 and Gazebo
- Implemented forward and inverse kinematics control pipelines
- Designed and built an autonomous firefighting robot (embedded control)
- Automated Excel workflows with Python, reducing processing time by 90%

Projects

SIMULATION OF A PICK-AND-PLACE ROBOT ARM | [GITHUB LINK](#) | FEB 2025

Technologies used: (ROS2, Gazebo, URDF, Python, Rviz)

- Modeled and validated a 7-DOF robotic manipulator in URDF using both primitive and mesh geometries, enabling accurate visualization and kinematic testing in RViz.
- Deployed physics-based simulation in Gazebo with ROS2 integration for real-time robot control.
- Developed custom ROS2 interfaces and messaging architecture enabling pose control, trajectory execution, and forward/inverse kinematics.
- Designed modular ROS2 packages with structured architecture, documentation, and maintainable code.

SELF-NAVIGATING FIREFIGHTING ROBOT | [GITHUB LINK](#) | JULY 2025

Technologies used: (C++, Arduino, Fritzing)

- Designed electrical architecture and selected sensing and actuation components using Fritzing.
- Built and integrated a fully autonomous mobile robot combining sensors, actuators, and onboard power systems.
- Implemented Arduino-based control logic for autonomous navigation, fire detection, and extinguishing

CONTROL OF A FOUR-DOF ROBOTIC ARM VIA BLUETOOTH | [GITHUB LINK](#) | AUGUST 2024

Technologies used: (C++, Arduino, MIT App inventor, TinkerCAD)

- Designed circuitry and assembled a 4-DOF robotic arm prototype using Arduino.
- Developed C++ control firmware enabling precise joint motion control.

- Built a mobile control interface with MIT App Inventor for wireless manipulation.
- Achieved low-latency real-time Bluetooth control with stable multi-joint operation.

Experience

PYTHON PROGRAMMER | INDEPENDENT | JAN 2022 – PRESENT

- Developed Python applications using APIs, data processing libraries, and automation workflows.
- Developed a smart literature review tool using Natural Language Processing
- Developed a 2-player tic-tac-toe game using the concepts of Object-Oriented-Programming and the pygame library.
- Developed a python script for the automation of Excel processes, cutting down processing time by 90% while minimizing error.

CHANNEL SALES INTERN | SCHNEIDER ELECTRIC | APRIL 2024 – SEPTEMBER 2024

- Supported installation and functional testing of industrial control panels, verifying wiring integrity and operational performance.
- Assisted channel partners by preparing technical product documentation and pricing datasets to support sales operations.
- Supported field engagement activities by demonstrating electrical products and explaining technical specifications to customers.

ENGINEERING INTERN | IKEJA ELECTRIC | JUNE 2023 – SEPTEMBER 2023

- Automated Excel workflows with Python (pandas, openpyxl), achieving a 90%-time reduction.
- Performed operational data analysis using Excel and Power BI to generate dashboards and identify performance trends.
- Performed electrical diagnostic tests including insulation resistance, winding resistance, and transformer turn ratio measurements to support maintenance evaluation.
- Analyzed electrical safety incidents and presented risk findings with recommended preventive measures.

Volunteer Experience

VOLUNTEER | ORILE-AGEGE GENERAL HOSPITAL AUGUST 2025

- Assisted medical staff with patient data recording and workflow coordination during community health outreach serving 2,500+ children.

Technical Skills

- Robotics: ROS2, Gazebo, RViz, MoveIt2
- Programming: Python, C++, Arduino
- Embedded Systems: ESP32, circuit design
- Data & Tools: Excel, Power BI, pandas